Grid-search protocol (v4) applied to the fBm-bank ESN with bounded non-parametric readouts and a modulated rough bank

22 hyperparameters searched (banks, modulation, numbers of readout pieces); the parameters on a randomised 2-level grid

Othmane Zarhali

October 19, 2026

Abstract

We apply the market-calibration protocol of the ESN–A2 note (v4) to the fBm-bank ESN with bounded non-parametric readouts and a modulated rough bank, written under $\mathbb{P}$:

$$\sigma_t = \tilde{\sigma}(t)\, e^{\psi(\zeta_t)} F(P_t), \qquad \tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt}),$$

with an initial state (ζ_0, P_0, Γ_0), bounded piecewise-linear readouts ψ (slopes c_m) and F (increments d_j, log-decays e_j), and a rough bank modulated by a Lévy-driven OU factor Γ.

- **Parameters**, gridded on a randomised 2-level design and never fitted: the readout values c, d, e, the level parameters a, b, c, the correlation ϱ and the initial state ζ_0, P_0, Γ_0.
- **Hyperparameters**, 22 of them, searched:
  - the rough bank $(M, H, \lambda_{lo}, \lambda_{hi})$;
  - the modulation $(\lambda_\Gamma, \Lambda, a_J)$;
  - the leverage window Θ;
  - the feedback bank $(J, \kappa_{lo}, \kappa_{hi}, w \in \mathbb{R}^J)$;
  - the numbers of readout pieces N_c, N_d, N_e.

From a cold start ($S_{\text{avg}} = 3.35$), the protocol's bounded LM and a Latin-hypercube stage stall at 2.12, with TRF ending on its step tolerance: the objective is piecewise constant in the integer hyperparameters. A derivative-free coordinate sweep followed by a last LM leg reaches $S_{\text{avg}} = 1.103$ after 579 evaluations, and the loss $S_{\text{avg}}(\theta_n) - S_{\text{avg}}(\theta^\star)$ falls from 2.25 to 0. Five seeds give 1.215 ± 0.078, against 1.066 ± 0.051 for the simpler model class of the earlier run.

The found architecture is:

- rough bank $M = 18$, $H = 0.058$, rates $[10^{-3}, 232]\,\text{yr}^{-1}$;
- modulation regimes with a half-life of about 71 days, 4.6 jumps per year;
- leverage window 37 days;
- a single fast feedback filter (about 3 days);
- readouts with $(N_c, N_d, N_e) = (5, 4, 1)$.

The (M, J) heat map has an interior minimum in J (at 2). The ablation makes three points:

- the modulation and the piecewise-linear readout shapes are essential;
- the counts of readout pieces beyond a minimal resolution, the gain side of F, the term structure $\tilde{\sigma}(t)$ and the initial state are within the noise;
- no single-coordinate change improves $\theta^\star$ beyond the 2σ floor.

Contents

1 Purpose and mapping of the protocol **2**

2 The model in continuous time **3**

3 Parameters and hyperparameters **5**

4 The grid-search algorithm, in detail **6**
4.1 Data and target references 6
4.2 The 11 smooth surrogate stylised facts 6
4.3 The parameter grid 7
4.4 One hyperparameter evaluation, end to end 8
4.5 The outer search 8
4.6 Score and loss 8

5 Simulation scheme **9**

6 Results **10**
6.1 Convergence and the loss 10
6.2 Found architecture 12
6.3 (M, J) sensitivity 14
6.4 Every stylised fact 15

7 Ablation **16**

8 Assessment **18**

A Code reference **19**
A.1 `gs_core.py`: model, simulation, statistics, score 19
A.2 Scripts 21

B Results files **21**

1 Purpose and mapping of the protocol

The protocol note (v4) defines a market-calibration procedure for a fixed model class, built on a split between two kinds of quantities:

- *hyperparameters* characterise the banks and the architecture, and are searched;
- *parameters* enter the volatility and the emitted return, and are averaged over a randomised 2-level grid, never fitted.

An earlier run applied the protocol to the fBm-bank ESN with a two-slope readout and the feedback $(1 + P^2)^\gamma$.

This document applies the protocol to a richer model class of the same family, whose components are:

- the term structure $\tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt})$;
- the initial state (ζ_0, P_0, Γ_0) at the start of every simulated window;
- bounded piecewise-linear readouts ψ (slopes c_m) and F (increments d_j, log-decays e_j);
- the Lévy-modulated rough bank with free $(\lambda_\Gamma, \Lambda, a_J)$;
- no drift.

The *numbers* of readout pieces N_c, N_d, N_e are hyperparameters, with the reference configuration $(N_c, N_d, N_e) = (6, 4, 3)$ (Lemma 2.2). The data, the 11 statistics, the targets, the score and the outer search are those of the earlier run, so the scores are directly comparable.

2 The model in continuous time

Time is in years ($\delta = 1/252$ per trading day). Every simulated window $[0, T_{\text{cal}}\delta]$ starts at $t = 0$. On $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$:

- $W^1, \dots, W^M, W^\perp$ are independent Brownian motions, and $W = (W^1, \dots, W^M)^\top$;
- L is an independent compound Poisson process with intensity Λ and $\text{Exp}(a_J)$ jump sizes.

Layer 0′: vol-of-vol modulation.

$$d\Gamma_t = -\lambda_\Gamma \Gamma_t \, dt + dL_t, \qquad \Gamma_0 > 0, \tag{2.1}$$

so that $\Gamma_t = \Gamma_0 e^{-\lambda_\Gamma t} + \int_0^t e^{-\lambda_\Gamma (t-s)} dL_s$, with stationary mean $\bar{\Gamma} = \Lambda/(a_J \lambda_\Gamma)$.

Layer 1: modulated rough bank. With $\lambda_i = \lambda_{lo}(\lambda_{hi}/\lambda_{lo})^{(i-1)/(M-1)}$, $i = 1, \dots, M$,

$$dr_t^i = -\lambda_i r_t^i \, dt + \sqrt{2\lambda_i \Gamma_t} \, dW_t^i, \qquad \zeta_t = \frac{q^\top r_t}{\sqrt{\gamma_0}}, \tag{2.2}$$

$$q_i = \sqrt{b_i}, \qquad b = \arg\min_{b \geq 0} \int_{\tau_{\min}}^{\tau_{\max}} \Big(\frac{2 \sum_i b_i (1 - e^{-\lambda_i \tau})}{\tau^{2H}} - 1 \Big)^2 \frac{d\tau}{\tau}, \qquad \gamma_0 = \sum_i b_i,$$

with fitting window $[\tau_{\min}, \tau_{\max}] = [1 \text{ day}, T_{\text{cal}}]$.

Layer 0: return noise and leverage.

$$dW_t^X = \varrho \, u^\top dW_t + \sqrt{1 - \varrho^2} \, dW_t^\perp, \qquad u = \frac{m}{\|m\|}, \quad m_i = q_i \sqrt{2\lambda_i} \, \frac{1 - e^{-\lambda_i \Theta}}{\lambda_i \Theta}. \tag{2.3}$$

Layer 2: endogenous feedback bank. For $j = 1, \dots, J$,

$$ds_t^j = -\kappa_j s_t^j \, dt + \sqrt{2\kappa_j} \, \frac{\sigma_t}{\tilde{\sigma}(t)} \, dW_t^X, \qquad P_t = w^\top s_t, \tag{2.4}$$

$$\kappa_j = \kappa_{hi} \Big(\frac{\kappa_{lo}}{\kappa_{hi}} \Big)^{\frac{j-1}{J-1}} \ (j = 1 \text{ fastest}), \qquad \kappa_1 = \sqrt{\kappa_{lo} \kappa_{hi}} \text{ if } J = 1, \qquad w \in \mathbb{R}^J, \ \|w\| = 1.$$

Readout, volatility and price.

$$\sigma_t = \tilde{\sigma}(t) \, e^{\psi(\zeta_t)} \, F(P_t), \qquad dX_t = -\tfrac{1}{2}\sigma_t^2 \, dt + \sigma_t \, dW_t^X, \qquad \tilde{\sigma}(t) = a \, e^{-bt} + c \, (1 - e^{-bt}). \tag{2.5}$$

Initial state. Given (ζ_0, Γ_0, P_0), with $g \sim \mathcal{N}(0, I_M)$,

$$r_0 = \sqrt{\Gamma_0} \, g - q \, \frac{q^\top \sqrt{\Gamma_0} \, g - \zeta_0 \sqrt{\gamma_0}}{\|q\|^2}, \qquad s_0 = P_0 \, w, \tag{2.6}$$

i.e. r_0 is drawn from the stationary law at level Γ_0, conditioned on $q^\top r_0 = \zeta_0 \sqrt{\gamma_0}$.

Measurement layer (protocol). Day n is $[t_n, t_{n+1}]$. It consists of an overnight interval of length $\phi\delta$ and a session S_n. The model emits

$$x_n = X_{t_{n+1}} - X_{t_n}, \qquad v_n = \tfrac{1}{2}\big(\sup_{S_n} X - \inf_{S_n} X\big)^2 - (2\ln 2 - 1)\big(X_{t_{n+1}} - X_{t_n+\phi\delta}\big)^2,$$

so the Garman–Klass proxy is applied to the model exactly as to the market.

Definition 2.1 (Readouts with N_c, N_d, N_e pieces)**.** 1. ψ**.** Continuous and piecewise linear on the $N_c + 1$ equispaced knots $z_k = -3 + 6k/N_c$ of $[-3, 3]$, with slope $-c_m \le 0$ on the m-th piece, normalised by $\psi(0) = 0$, and constant beyond ± 3.

2. F**.** Continuous and piecewise linear on the knots $p_{-N_d} < \cdots < p_{-1} < p_0 = 0 < p_1 < \cdots < p_{N_e}$, with

$$F(0) = 1, \qquad F(p_{-k}) = F(p_{-k+1}) + d_{N_d-k+1} \quad \text{(loss side, } d \ge 0\text{)},$$
$$F(p_k) = F(p_{k-1})\, e^{-e_k} \quad \text{(gain side, } e \ge 0\text{)},$$

and constant beyond the outer knots. The knot positions are

$$p_{-k} = -\ell_-\Big(\frac{k}{N_d}\Big), \qquad p_k = \ell_+\Big(\frac{k}{N_e}\Big),$$

where ℓ_- is the piecewise-linear map through $(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1) \mapsto (0, 1.5, 3, 5, 8)$ and ℓ_+ the one through $(0, \frac{1}{3}, \frac{2}{3}, 1) \mapsto (0, 1.5, 3, 5)$.

Lemma 2.2 (Reference configuration $(N_c, N_d, N_e) = (6, 4, 3)$)**.** *For $N_c = 6$ the ψ-knots are the integers $-3, -2, \ldots, 3$. For $(N_d, N_e) = (4, 3)$ the F-knots are $-8, -5, -3, -1.5, 0, 1.5, 3, 5$. For general counts, the knots refine or coarsen this configuration along the same maps $\ell_\pm$.*

Proof. $z_k = -3 + k$ for $N_c = 6$. Evaluating ℓ_- at $k/4$, $k = 1, \ldots, 4$, gives $1.5, 3, 5, 8$, and evaluating ℓ_+ at $k/3$ gives $1.5, 3, 5$. □

Proposition 2.3 (Boundedness at every grid point)**.** *Under Definition 2.1,*

$$\sigma_t \le \max(a, c)\, e^{3\max_m c_m}\, \Big(1 + \sum_j d_j\Big).$$

Hence σ, X, P and the measured (x_n, v_n) have finite moments of all orders, and the system (2.1)–(2.6) *has a unique strong solution.*

Proof. $\tilde\sigma(t)$ is a convex combination of a and c. Since $\psi(0) = 0$, ψ is non-increasing and constant beyond -3, we have $\psi \le \psi(-3) \le 3\max_m c_m$. Likewise F is non-increasing and constant left of p_{-N_d}, so $F \le F(p_{-N_d}) = 1 + \sum_j d_j$. For existence and uniqueness: Γ is explicit and positive, and r solves a linear SDE driven by $\sqrt{\Gamma}$. Given (Γ, r), the pair (s, X) solves an SDE whose coefficients are bounded and Lipschitz in s, since ψ and F are piecewise linear and bounded, and so is $1/\tilde\sigma$. □

Note. On the parameter boxes of Table 1 the bound is at most $0.25\, e^{2.1}(1 + 1.2) = 4.5$, i.e. 450% annualised. No grid corner can explode, whatever the counts (N_c, N_d, N_e) and the other hyperparameters.

Remark 2.4 (The modulation in a time series). Γ is a common random variance of all rough factors. Given its path, ζ_t is centred Gaussian with variance

$$\gamma_0^{-1} \sum_i b_i \Big(e^{-2\lambda_i t}\, \Gamma_0 + \int_0^t 2\lambda_i e^{-2\lambda_i (t-v)} \Gamma_v\, dv\Big),$$

which mixes the readout ψ over a random scale. Bursts of Γ (rate Λ, mean size $1/a_J$, lifetime $1/\lambda_\Gamma$) therefore act on the tail statistics (99.5% quantile and maximum of volatility, kurtosis) and on volatility persistence.

Proposition 2.5 (J and w set the amplitude of the feedback signal)**.** *All feedback states are driven by the same* $(\sigma/\tilde{\sigma})dW^X$*. If* $\kappa_1 = \cdots = \kappa_J$ *and* $s_0 = P_0 w$*, then* $s_t^j = w_j P_0 e^{-\kappa t} + \eta_t$ *with a common* η*, and*

$$P_t = P_0 e^{-\kappa t} + \Big(\sum_j w_j\Big)\eta_t, \qquad \Big|\sum_j w_j\Big| \le \sqrt{J} \quad (\|w\| = 1).$$

For distinct rates, $\operatorname{Var} P_t = \sum_{j,k} w_j w_k \operatorname{Cov}(s_t^j, s_t^k)$ *with positive covariances. The amplitude of* P *decides which knots of* F *are visited.*

Proof. The equations are linear in s with a common driver; split the solution into the decaying initial term and the stochastic integral. Then apply Cauchy–Schwarz, and use $\operatorname{Cov}(s^j, s^k) \approx 2\sqrt{\kappa_j \kappa_k}/(\kappa_j + \kappa_k)\,\mathbb{E}[(\sigma/\tilde{\sigma})^2] > 0$ in stationarity. □

3 Parameters and hyperparameters

Definition 3.1 (The split)**.** *Parameters* are the quantities of (2.5) and (2.6), i.e. everything that is not a bank object:

$$\theta_P = \big(c_1, \ldots, c_{N_c},\ d_1, \ldots, d_{N_d},\ e_1, \ldots, e_{N_e},\ a,\ b,\ c,\ \varrho,\ \zeta_0,\ P_0,\ \Gamma_0\big),$$

with $N_c + N_d + N_e + 7$ coordinates (20 in the reference configuration). *Hyperparameters* define the banks and the architecture:

$$\theta_H = \big(M, H, \lambda_{lo}, \lambda_{hi};\ \lambda_\Gamma, \Lambda, a_J;\ \Theta;\ J, \kappa_{lo}, \kappa_{hi}, w;\ N_c, N_d, N_e\big).$$

These are the rough bank, the modulation, the leverage window, the feedback bank and the numbers of readout pieces. *Fixed*:

- the knot rule of Definition 2.1;
- $\delta = 1/252$ and $[\tau_{\min}, \tau_{\max}]$;
- the overnight share $\phi = 0.168$ (mean of the per-asset values 0.105, 0.227, 0.313, 0.137, 0.057);
- 13 sub-steps per day.

Remark 3.2 (Why N_c, N_d, N_e are hyperparameters). They define the architecture of the readouts: the dimension of θ_P, the knots, and which values of ζ and P are resolved. The values c, d, e placed on these knots enter σ directly and are parameters. This mirrors the bank sizes M and J: hyperparameters, whose rates and loadings define the objects that the parameters read.

Encoding. The search vector has 22 coordinates:

- log-encoded: $M, \lambda_{lo}, \lambda_{hi}, \lambda_\Gamma, \Lambda, a_J, \Theta, J, \kappa_{lo}, \kappa_{hi}, N_c, N_d, N_e$; the integers M, J, N_c, N_d, N_e are rounded after decoding;
- direct: H and $w^{\text{raw}} \in [-1, 1]^8$, with $w = w^{\text{raw}}_{1:J}/\|w^{\text{raw}}_{1:J}\|$.

Table 1: Parameters: the two levels of the randomised design (averaged, never fitted). The d and e levels scale with $1/N_d$ and $1/N_e$, so that the total amplitude of F at a corner does not depend on the number of pieces.

parameter	low	high	meaning
c_m, piece left of 0	0.3	0.7	slope of $-\psi$ for $\zeta < 0$
c_m, piece right of 0	0.2	0.6	slope of $-\psi$ for $\zeta > 0$
c_m, piece centred at 0	0.25	0.65	(N_c odd)
d_j	0	$1.2/N_d$	loss-side increments of F; $F(p_{-N_d}) \le 2.2$
e_j	0	$0.9/N_e$	gain-side log-decays of F; $F(p_{N_e}) \ge 0.41$
a	0.15	0.25	level of $\tilde{\sigma}$ at $t = 0$
b (yr^{-1})	0.5	5	speed of $\tilde{\sigma}$
c	0.15	0.25	long-run level of $\tilde{\sigma}$
ϱ	0.2	0.7	leverage correlation
ζ_0	-1.5	1.5	rough state at $t = 0$
P_0	-1.5	1.5	feedback state at $t = 0$
Γ_0	0.3	1.5	modulation at $t = 0$

Note. The slope boxes are those of s_-, s_+ in the earlier run, now applied piece by piece, and the level boxes are those of $\tilde{\sigma}$. The levels of ζ_0, P_0, Γ_0 cover typical states at the start of a window.

4 The grid-search algorithm, in detail

4.1 Data and target references

Five S&P 500 tickers are taken from `list_dics_SP500_ohlcvol_19940101_filtered.pkl`: **GPC, QCOM, INTC, LUV, NI**. For each ticker:

- the last $8\,052 = 6 \times 1\,342$ trading days are split into $N_{\text{win}} = 6$ windows of $T_{\text{cal}} = 1\,342$ days;
- on each window, the daily log-return $x_t = \log(C_t/C_{t-1})$ and the Garman–Klass proxy

$$v_t = \tfrac{1}{2}\log^2(H_t/L_t) - (2\ln 2 - 1)\log^2(C_t/O_t), \qquad v_t \leftarrow \max\big(v_t, 10^{-3}\text{med}(v)\big),$$

  feed the 11 statistics of Section 4.2.

The centre and tolerance of asset a and fact k are

$$c_k^{(a)} = \frac{1}{N_{\text{win}}}\sum_{\ell} s_k^{(a,\ell)}, \qquad s_k^{(a)} = \max\Big(\text{sd}_\ell\big(s_k^{(a,\ell)}\big),\ 0.30|c_k^{(a)}|,\ \text{floor}_k\Big),$$

with floors $(0.005, 0.01, 0.03, 0.05, 0.02, 0.005, 0.03, 0.01, 0.005, 0.3, 0.01)$.

4.2 The 11 smooth surrogate stylised facts

Notation.

- $sd_t = \sqrt{v_t}$ and ann $= \sqrt{252}$;
- abs_acf(L), sq_acf(L) and ret_acf(L) are the lag-L autocorrelations of $|x_t|$, x_t^2 and x_t, for $L = 1, \dots, 20$.

1. $\hat{H}$**.** With $L_1 < \cdots < L_{15}$ log-spaced in $[1, n/4]$, $d(L) = \overline{(\log v_{t+L} - \log v_t)^2}$ and $\hat{H} = \frac{1}{2}\,\text{slope}(\log L, \log d(L))$.
2. **Mean annualised volatility**: ann $\cdot\ \overline{sd_t}$.
3. **99.5% quantile (Harrell–Davis).** The weights are $w_i = I_{i/n}(a,b) - I_{(i-1)/n}(a,b)$, with $a = (n+1)q$, $b = (n+1)(1-q)$ and $q = 0.995$. The statistic is ann $\sum_i w_i sd_{(i)}$.

4. **Maximum annualised volatility (soft-max).** The soft-max is

$$m + \beta^{-1} \log\Big(\frac{1}{n}\sum_t e^{\beta(sd_t - m)}\Big),$$

with $\beta = 8/\hat{\sigma}_{sd}$ and $m = \max_t sd_t$; the statistic is ann times it.

5. **Mean volatility ACF**: $\frac{1}{20}\sum_L \mathrm{corr}(v_t, v_{t+L})$.

6. **Taylor gap**: $\overline{\text{abs_acf}} - \overline{\text{sq_acf}}$.

7. **Taylor fraction**: $\frac{1}{20}\sum_L \big(1 + e^{-200[\text{abs_acf}(L) - \text{sq_acf}(L)]}\big)^{-1}$.

8. **Zumbach.** The mean over $L \in \{5, 10, 20\}$ of $\mathrm{corr}(p_R^2, f_V) - \mathrm{corr}(p_V, f_R^2)$, where p and f denote past and future L-day sums of returns (R) and mean v (V).

9. **Leverage**: $\frac{1}{20}\sum_L \mathrm{corr}(x_t, v_{t+L})$.

10. **Excess kurtosis (tanh-winsorised).** With $c_t = x_t - \bar{x}$ and $c_s = 12\sqrt{\mathrm{Var}\, x}$, the statistic is

$$\overline{(c_s \tanh(c_t/c_s))^4} / \mathrm{Var}(x)^2 - 3.$$

11. **Maximum |return ACF|**: the soft-max of $\{|\text{ret_acf}(L)|\}_{L \le 20}$.

4.3 The parameter grid

The number of parameters depends on (N_c, N_d, N_e). The design is therefore defined on *slots*: 12 c-slots, 8 d-slots, 8 e-slots and 7 scalar slots, 35 in all.

1. A fixed seed (2026) draws a 20×35 matrix of i.i.d. Bernoulli($\frac{1}{2}$) bits $\beta_{g,\ell}$.

2. The active parameter in slot ℓ takes, at grid point g, its high level if $\beta_{g,\ell} = 1$ and its low level otherwise (Table 1).

3. Inactive slots ($m > N_c$, $j > N_d$, $k > N_e$) are ignored.

Lemma 4.1 (Stability of the design across readout counts)**.** *Changing (N_c, N_d, N_e) does not change the levels of the parameters common to both configurations, nor the common random numbers. Two hyperparameter points that differ only in the counts are therefore compared on the same scalar parameters and the same noise.*

Proof. The bit of a parameter depends only on its slot, and the slot of c_m (d_j, e_k, a scalar) does not depend on the counts. The random streams are indexed by path and time only. □

Each grid point is simulated with $N_{\text{cal}} = 8$ paths of T_{cal} days.

4.4 One hyperparameter evaluation, end to end

Algorithm 1 One call to `Residuals`(θ_H): 22-dimensional $\theta_H \to$ 55-entry residual

decode $\theta_H \in \mathbb{R}^{22}$ (13 log-encoded entries; M, J, N_c, N_d, N_e rounded; H, w^{raw} direct; all box-clipped, Table 3) $\to$ dict h
`bank`$\leftarrow$`build_bank`(h): $\lambda_{1:M}$, NNLS weights b, $q = \sqrt{b}$, γ_0; $u(\Theta)$; $\kappa_{1:J}$, w; $(\lambda_\Gamma, \Lambda, a_J)$; knots of ψ and F ▷ built ONCE
$G \leftarrow$ `param_grid`(N_c, N_d, N_e): levels of the $N_c + N_d + N_e + 7$ parameters at the 20 grid points (slot design)
per path: knot values of ψ (cumulated slopes, $\psi(0) = 0$) and F (cumulated increments and decays)
simulate in one vectorised pass $20 \times 8 = 160$ paths of T_{cal} days from the initial state (2.6) (Section 5) ▷ CRN: seeds 11 and $11 + 7919$
for $g = 1, \ldots, 20$ **do**
    compute the 11 statistics on each of the 8 paths; $\bar{s}^{(g)} \leftarrow$ their mean
end for
$\bar{s} \leftarrow \frac{1}{20} \sum_g \bar{s}^{(g)}$ ▷ an average of averages
for each asset a and $k = 1, \ldots, 11$ **do**
    $z_k^{(a)} \leftarrow (\bar{s}_k - c_k^{(a)})/s_k^{(a)}$; $r_k^{(a)} \leftarrow \sqrt{w_k} z_k^{(a)}$
end for
return $r \in \mathbb{R}^{55}$; log $(\theta_H, h, Q = \|r\|^2, S_{\text{avg}}, \bar{s})$ in the trace

4.5 The outer search

Algorithm 2 Outer search over θ_H

Require: box bounds (Table 3) in encoded coordinates; `max_nfev`$= 15$ per leg
Stage A (cold-start LM). $\theta_0 \leftarrow$ centre of the box (geometric for log-encoded entries; $w^{\text{raw}} = 0.5$); bounded TRF, `diff_step`$= 0.1$
Stage B (global exploration, in parallel with A). 48-point Latin hypercube in the encoded box (seed 7), each point evaluated by Alg. 1
Stage C (LM legs).
repeat
    $\theta_{\text{start}} \leftarrow \arg\min_{\text{trace}} Q$ over all stages
    bounded TRF from θ_{start}; forward-difference Jacobian with the steps of scipy's 2-point scheme, its 22 columns evaluated on 2 processes
until a leg improves the best S_{avg} by less than the 5-seed noise, or TRF stops
Stage D (coordinate sweep). From $\arg\min_{\text{trace}} Q$, for 2 passes and each active coordinate i in turn: evaluate the $K = 7$ equispaced values of the encoded box in coordinate i, all others fixed (in parallel), and adopt the best if it lowers Q
Stage C3. One more bounded TRF leg from $\arg\min_{\text{trace}} Q$
return $\theta^\star = \arg\min_{\text{trace}} Q$

Note. The integer hyperparameters are searched in continuous encoded coordinates and rounded on decoding, so the objective is piecewise constant in them; it is also piecewise constant in Λ. A finite-difference step of 10% in log-coordinates moves N_c, N_d, N_e by one unit only when the current value lies near a rounding boundary. In this run, LM legs C1 and C2 ended on the step tolerance (TRF status 3) while the ablation still exhibited descent directions (Section 6.1). Stage D is derivative-free and moves every coordinate, integers included, across its whole box. It is the addition to the protocol's outer search required by this model class.

4.6 Score and loss

For asset a,

$$S^{(a)} = \sum_{k=1}^{11} w_k \big(1 - e^{-\frac{1}{2}(z_k^{(a)})^2}\big) + 5 \cdot \text{Stress}, \qquad S_{\text{avg}} = \frac{1}{5} \sum_a S^{(a)},$$

with weights $w = (2.2, 1.4, 0.6, 0.2, 0.8, 1.0, 0.8, 1.0, 1.1, 0.7, 0.5)$. Stress $= 1$ if the grid-averaged mean annualised volatility exceeds 1.5 or a statistic is not finite. TRF minimises $Q = \|r\|^2$, and S_{avg} is reported.

S_{avg} has a positive floor, set by three effects: corner averaging, each asset's deviation from the five-asset compromise, and Monte Carlo noise. Convergence is therefore reported through the **loss**

$$L_n = S_{\text{avg}}(\theta_n) - S_{\text{avg}}(\theta^\star), \qquad \theta^\star = \arg\min_m Q(\theta_m),$$

whose running minimum starts high at the cold start and drops to 0 at $\theta^\star$.

5 Simulation scheme

Each day has 13 sub-steps of length $h = \delta/13$. σ and Γ are frozen within a sub-step, and $\tilde{\sigma}$ is evaluated at the start of the sub-step. The paths start at $t = 0$ from (2.6), with **no burn-in**: the window starts at the initial state, as in the model. Per sub-step:

1. **Readout.** $\zeta = q^\top r/\sqrt{\gamma_0}$, $P = w^\top s$, $\sigma = \tilde{\sigma}(t)e^{\psi(\zeta)}F(P)$, with ψ and F interpolated row-wise from the per-path knot values.

2. **Layers 1 and 0.** Draw the exact bivariate law of

$$\Big(\Delta W^i,\ \int \sqrt{2\lambda_i}e^{-\lambda_i(\cdot)}dW^i\Big),$$

   with variances h and $1 - e^{-2\lambda_i h}$ and covariance $\sqrt{2\lambda_i}(1 - e^{-\lambda_i h})/\lambda_i$. The OU noise is scaled by $\sqrt{\Gamma}$. Then $\Delta W^X = \varrho u^\top \Delta W + \sqrt{1-\varrho^2}\Delta W^\perp$.

3. **Price.** $X \leftarrow X - \frac{1}{2}\sigma^2 h + \sigma \Delta W^X$.

4. **Layer 2.** $s^j \leftarrow e^{-\kappa_j h}s^j + \sqrt{1 - e^{-2\kappa_j h}}\,(\sigma/\tilde{\sigma}(t))\,\Delta W^X/\sqrt{h}$.

5. **Layer 0′.** $\Gamma \leftarrow e^{-\lambda_\Gamma h}\Gamma + \mathbf{1}\{U < 1 - e^{-\Lambda h}\}Z$, with $U \sim \mathcal{U}(0,1)$ and $Z \sim \text{Exp}(a_J)$ from a separate random stream. At most one jump per sub-step is allowed; the probability of two is at most $(\Lambda h)^2/2 \le 4 \times 10^{-5}$.

6. **Measurement.** The first round$(13\phi) = 2$ sub-steps of a day are overnight. In the session, the extremes of the Brownian bridge are sampled exactly, as $\frac{a+b}{2} \pm \frac{1}{2}\sqrt{(b-a)^2 - 2\sigma^2 h \log U}$.

Lemma 5.1 (Initial rough state)**.** *Let $g \sim \mathcal{N}(0, I_M)$, $\Gamma_0 > 0$, and r_0 as in (2.6). Then $q^\top r_0 = \zeta_0\sqrt{\gamma_0}$, and r_0 has the law of $\mathcal{N}(0, \Gamma_0 I_M)$ conditioned on this linear constraint.*

Proof. $q^\top r_0 = q^\top\sqrt{\Gamma_0}g - (q^\top\sqrt{\Gamma_0}g - \zeta_0\sqrt{\gamma_0}) = \zeta_0\sqrt{\gamma_0}$. For an isotropic Gaussian vector $y = \sqrt{\Gamma_0}g$, the orthogonal projection $y - q\,q^\top y/\|q\|^2$ is independent of $q^\top y$ and has the conditional law of y given $q^\top y$, minus its conditional mean. Adding back $q\,\zeta_0\sqrt{\gamma_0}/\|q\|^2$ gives the conditional law at the prescribed value. □

Note. The Gaussian stream (seed 11) and the jump stream (seed $11 + 7919$) are separate generators. Changing $(\lambda_\Gamma, \Lambda, a_J)$ therefore does not shift the noise of the other layers, which keeps the common random numbers effective for the finite-difference Jacobian. The objective is piecewise constant in Λ (a jump occurs when $U < 1 - e^{-\Lambda h}$) and smooth in a_J and λ_Γ.

6 Results

6.1 Convergence and the loss

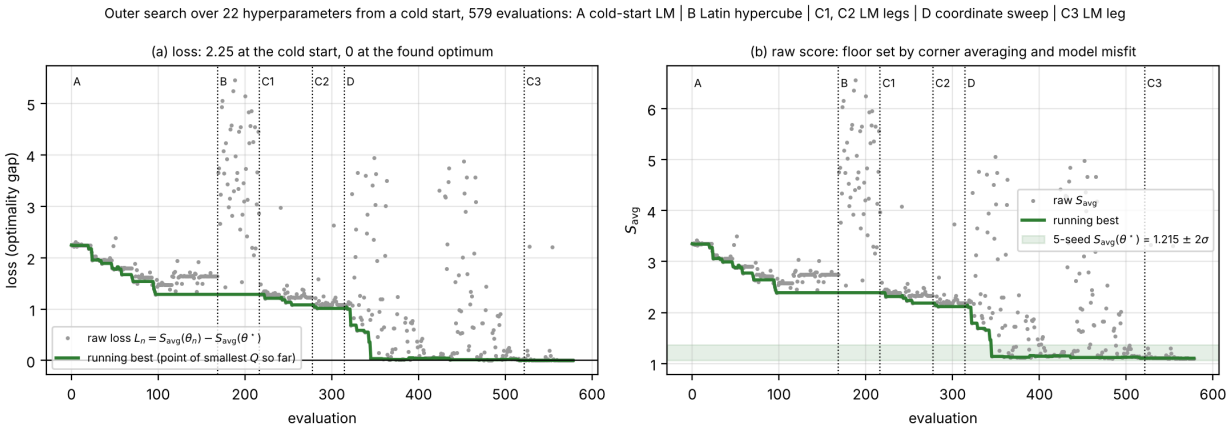


Figure 1: (a) Loss $L_n = S_{\text{avg}}(\theta_n) - S_{\text{avg}}(\theta^\star)$ and its running best, from 2.25 at the cold start to 0 at $\theta^\star$. (b) Raw S_{avg}, with the 5-seed band at $\theta^\star$. 579 evaluations in six stages.

Table 2: Stages of the outer search (Algorithm 2).

stage	evaluations	first S_{avg}	best S_{avg}
A: cold-start LM	0–168	3.348	2.388
B: Latin hypercube	169–216	4.752	2.635
C1: LM leg	217–277	2.388	2.184
C2: LM leg	278–314	2.184	2.116
D: coordinate sweep	315–521	2.127	1.105
C3: LM leg	522–578	1.105	1.103

1. **Stage A.** The protocol's bounded LM, started at the centre of the box, descends from 3.35 to 2.39 and stalls.

2. **Stage B.** The 48 Latin-hypercube points span $S_{\text{avg}} \in [2.53, 6.56]$. None beats stage A in Q; the best of them by Q scores 2.64.

3. **Stages C1 and C2.** The two LM legs reach 2.18 and then 2.12, and both end on the step tolerance (TRF status 3). The improvement of C2 (0.07) is below the 5-seed noise, so the stopping rule of the original protocol is met. Yet an ablation run at that point showed that a single coordinate move ($\Theta \to 5$ days) still lowered S_{avg} by 0.31, more than 2σ. The point was not a local minimum: the LM stalled because the objective is piecewise constant in the five integer hyperparameters and in Λ, and noisy under common random numbers.

4. **Stage D.** The derivative-free coordinate sweep makes 19 accepted moves in two passes and lowers S_{avg} from 2.12 to 1.105. The largest gains come from H ($0.14 \to 0.058$, Q from 26.8 to 22.4) and from the modulation rate λ_Γ ($4.4 \to 2.5\,\text{yr}^{-1}$, Q from 20.7 to 13.9). Then Θ, J, κ and the readout counts follow.

5. **Stage C3.** A last LM leg from the swept point improves by 0.002 and stops on the step tolerance.

At $\theta^\star$, every single-coordinate ablation of Section 7 is either worse or within the $2\sigma = 0.156$ noise floor. $\theta^\star$ is therefore a local minimum at the resolution of the noise. The loss falls from 2.25 to 0.

The raw score has a floor: five seeds at $\theta^\star$ give

$$S_{\text{avg}} = 1.215 \pm 0.078 \qquad (1.103,\ 1.197,\ 1.306,\ 1.270,\ 1.202).$$

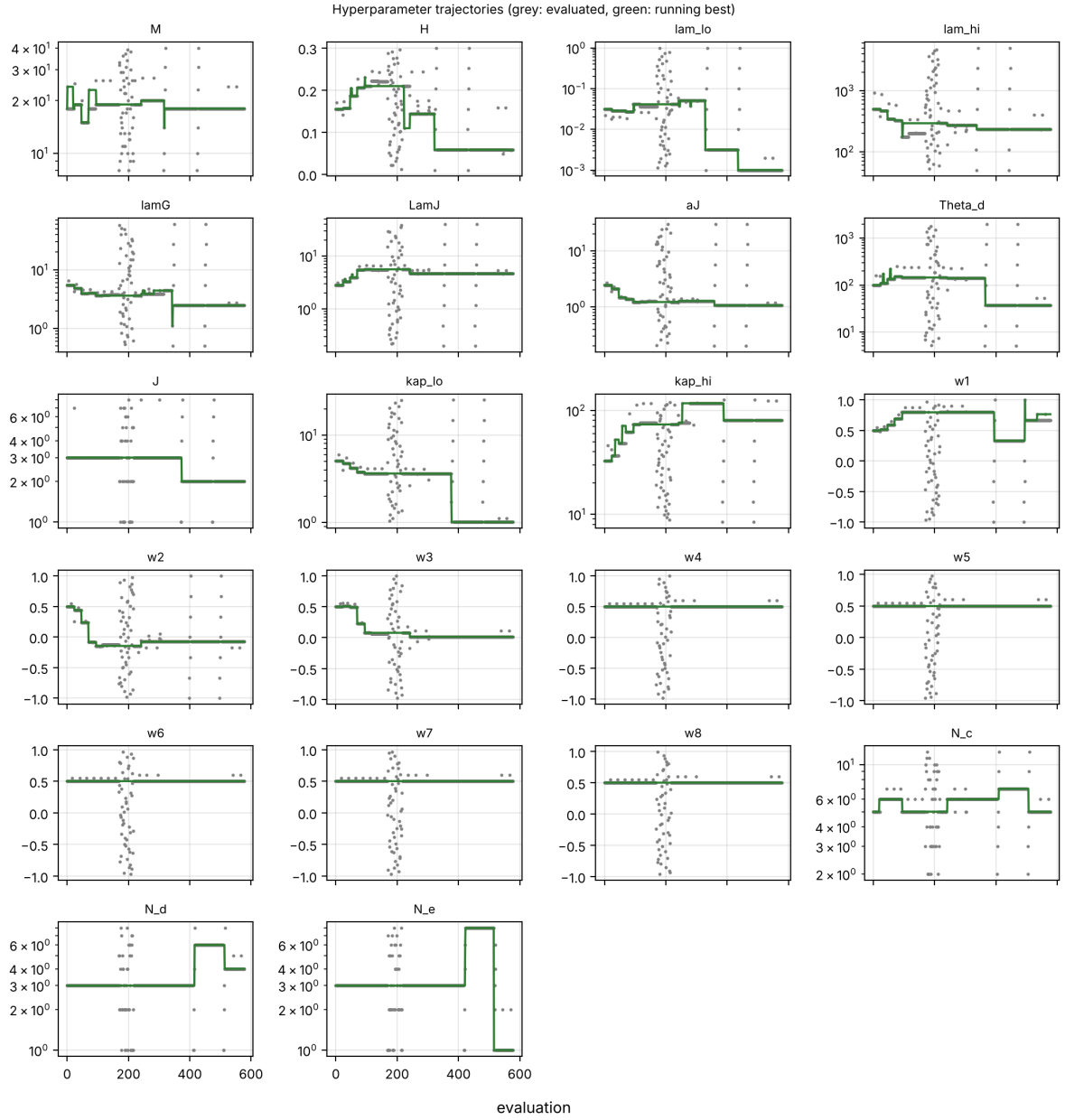


Figure 2: The 22 hyperparameter trajectories (grey: evaluated; green: point of smallest Q so far). The jumps of the running best in stage D are the coordinate moves.

6.2 Found architecture

Table 3: Hyperparameter box and the found optimum $\theta^\star$ (evaluation 561 of 579).

hyperparameter	low	high	found
M (rough modes)	8	40	18
H	0.01	0.3	0.05773
λ_{lo} (yr^{-1})	0.001	1	0.001 (bound)
λ_{hi} (yr^{-1})	50	5000	231.9
λ_Γ (yr^{-1})	0.5	60	2.467
Λ (yr^{-1})	0.2	40	4.628
a_J	0.2	30	1.063
Θ (days)	5	2000	36.84
J (feedback states)	1	8	2
κ_{lo} (yr^{-1})	1.008	25.2	1.008 (bound)
κ_{hi} (yr^{-1})	8.4	126	80.22
w_1^{raw}	-1	1	0.7667 ($w_1 = 0.995$)
w_2^{raw}	-1	1	-0.07592 ($w_2 = -0.099$)
w_3^{raw}	-1	1	0.0133 (inactive, $j > J$)
w_4^{raw}	-1	1	0.5 (inactive, $j > J$)
w_5^{raw}	-1	1	0.5 (inactive, $j > J$)
w_6^{raw}	-1	1	0.5 (inactive, $j > J$)
w_7^{raw}	-1	1	0.5 (inactive, $j > J$)
w_8^{raw}	-1	1	0.5 (inactive, $j > J$)
N_c (pieces of ψ)	2	12	5
N_d (loss-side pieces of F)	1	8	4
N_e (gain-side pieces of F)	1	8	1

1. **Rough bank.** $H = 0.0577$ and $\lambda_{hi} = 232\,\mathrm{yr}^{-1}$ almost coincide with the earlier optimum (0.0596 and 235). With λ_{lo} at its lower bound $10^{-3}\,\mathrm{yr}^{-1}$, the 18 modes now span five decades: the bank is long-memory as well as rough. The NNLS variogram fit has a maximal relative error of 24%, and $\gamma_0 = 0.679$.

2. **Modulation.** The search selects:

   - $\lambda_\Gamma = 2.47\,\mathrm{yr}^{-1}$, a half-life of 71 trading days;
   - $\Lambda = 4.63$ jumps per year, of mean size $1/a_J = 0.94$;
   - stationary mean $\bar{\Gamma} = \Lambda/(a_J\lambda_\Gamma) = 1.76$.

   These are vol-of-vol regimes lasting about a quarter, several per year (Fig. 5).

3. **Leverage window.** $\Theta = 36.8$ days, against about 10 days earlier; the ablation shows a flat valley between 5 and 37 days.

4. **Feedback bank.** $J = 2$, with time scales of 3.1 and 250 days (κ_{lo} at its bound). The weights are $w = (0.995, -0.099)$: in effect a single fast filter, as the ablation "$w \to$ fastest filter only" ($\Delta = 0.000$) confirms.

5. **Readout counts.** $(N_c, N_d, N_e) = (5, 4, 1)$. ψ has five pieces with knots -3, -1.8, -0.6, 0.6, 1.8, 3. F keeps the four reference loss-side knots -8, -5, -3, -1.5 and a single gain-side piece to 5.

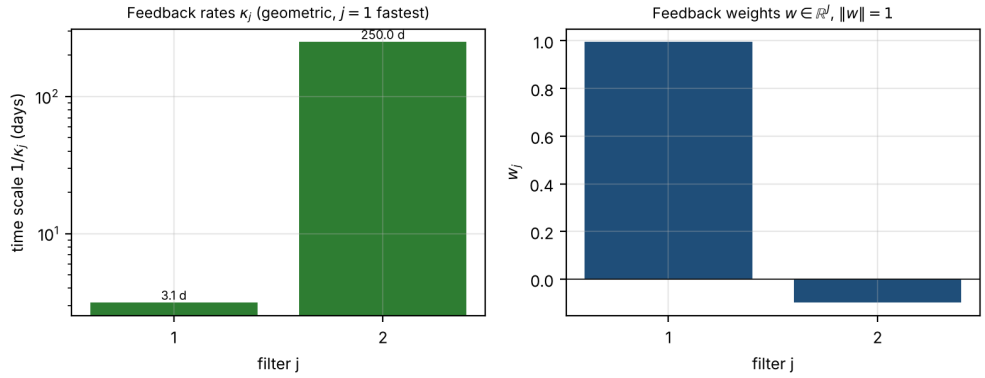


Figure 3: The found feedback bank: $J = 2$, time scales 3.1 and 250 days; the weight is concentrated on the fast filter.

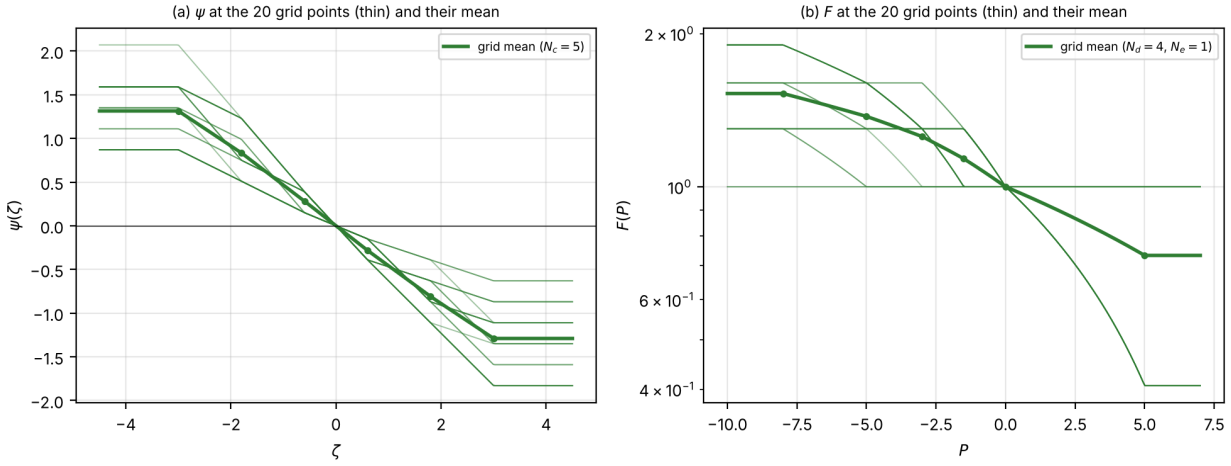


Figure 4: Readouts at $\theta^\star$: ψ ($N_c = 5$) and F ($N_d = 4$, $N_e = 1$) at the 20 grid points (thin) and their grid mean (thick).

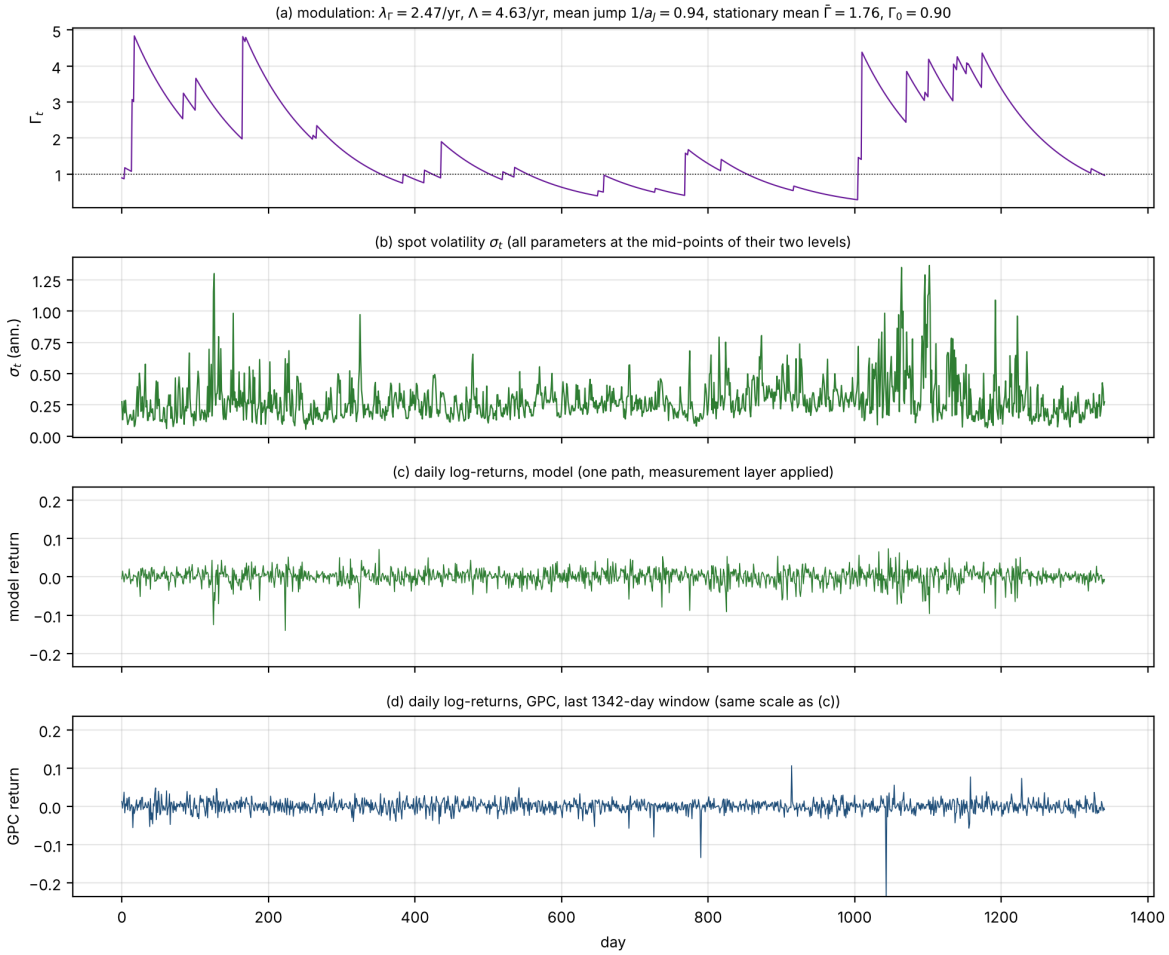


Figure 5: One simulated path at $\theta^\star$ with all parameters at the mid-points of their levels: (a) Γ_t; (b) σ_t; (c) daily returns after the measurement layer; (d) GPC returns over its last window, on the same scale.

6.3 (M, J) sensitivity

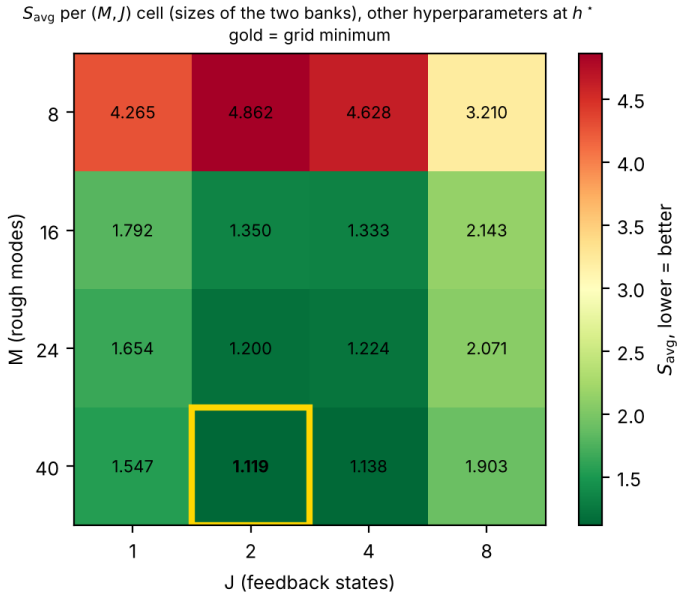


Figure 6: S_{avg} on a 4×4 grid of the two bank sizes, other hyperparameters at $\theta^\star$ (seed 11).

- J**.** For every $M \geq 16$, $J = 2$ is best, with $J = 4$ within noise. $J = 1$ and $J = 8$ are clearly worse, so the minimum in J is interior. $J = 8$ over-amplifies the feedback (Proposition 2.5).
- M**.** $M = 8$ is catastrophic (3.2–4.9). With $\lambda_{lo} = 10^{-3}$, eight modes cannot cover five decades of rates and match the fBm variogram. From $M = 16$ to 40 the score improves slowly, from 1.35 to 1.12 at $J = 2$. The found $M = 18$ scores 1.103 on the search seed, below every cell, because the other hyperparameters were optimised jointly with it.

6.4 Every stylised fact

Table 4: Window-mean targets per asset; the grid-averaged found model (seed 11); and, for comparison, the optimum of the earlier run (two-slope ψ, $(1 + P^2)^\gamma$, unmodulated bank).

stylised fact (w_k)	GPC	QCOM	INTC	LUV	NI	**found**	earlier run
$\hat{H}$ (2.2)	0.0495	0.0533	0.0561	0.0463	0.0542	**0.0563**	0.0549
mean ann. vol (1.4)	0.1846	0.3204	0.2695	0.2863	0.1779	**0.2197**	0.2336
$q_{99.5}$ ann. vol (0.6)	0.6977	1.1509	0.8994	1.0409	0.7149	**0.8582**	0.8413
max ann. vol (0.2)	1.0315	1.7909	1.4867	1.4971	1.1223	**1.1249**	1.1574
mean vol ACF (0.8)	0.2158	0.1869	0.2356	0.1961	0.2253	**0.1734**	0.1795
Taylor gap (1)	0.0388	0.0477	0.0478	0.0368	0.0466	**0.0348**	0.0268
Taylor fraction (0.8)	0.8473	0.8958	0.9542	0.8492	0.8707	**0.8304**	0.7691
Zumbach (1)	0.1081	0.1291	0.1293	0.1064	0.1033	**0.0557**	0.0938
leverage (1.1)	−0.0340	−0.0308	−0.0357	−0.0323	−0.0403	**−0.0432**	−0.0320
excess kurtosis (0.7)	5.576	4.474	4.567	3.661	4.979	**4.168**	3.220
max|ret ACF| (0.5)	0.0941	0.0775	0.0945	0.0793	0.0813	**0.0679**	0.0660

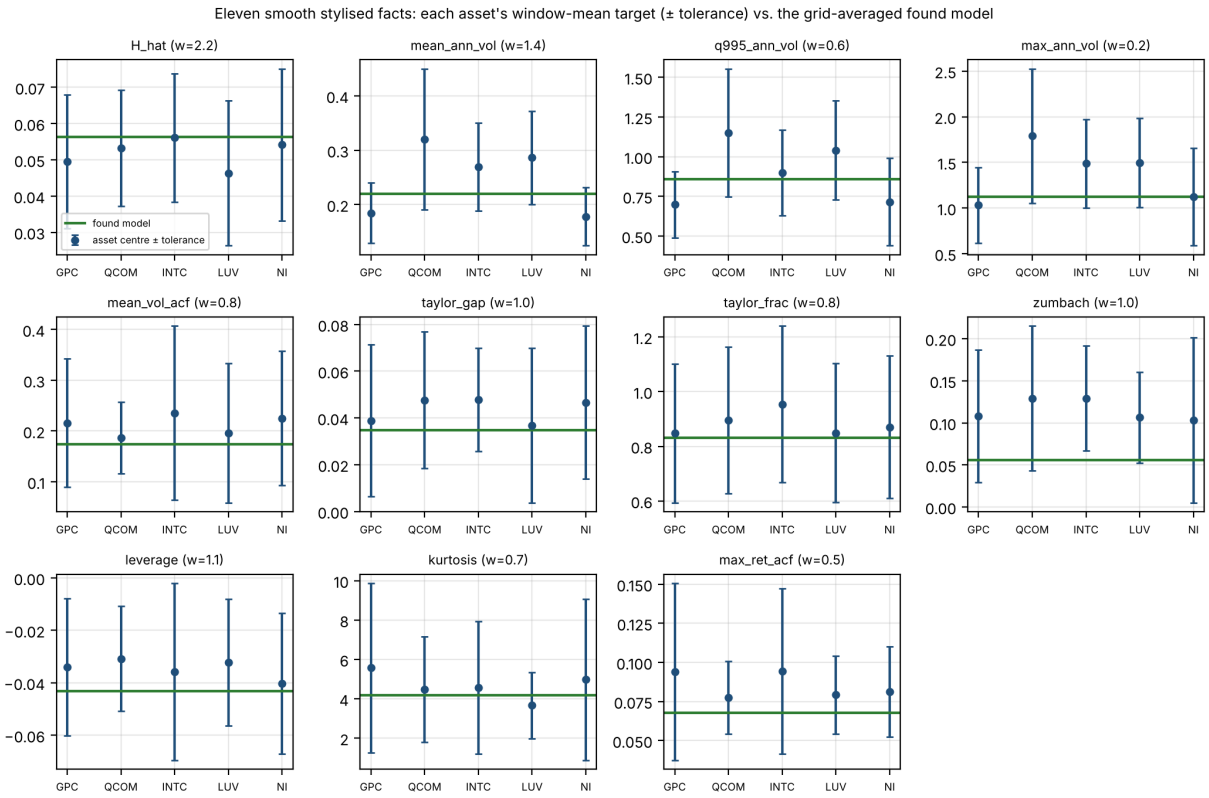


Figure 7: Each asset's target (± tolerance) against the grid-averaged found model.

- **Within the five assets' range:** $\hat{H}$ (0.056), mean annualised volatility (0.220), $q_{99.5}$ (0.858), maximum volatility (1.125), Taylor fraction (0.830, just below), Taylor gap (0.035) and excess kurtosis (4.17).

- **Improved over the earlier run:** kurtosis (4.17 against 3.22, now inside the 3.66–5.58 range), Taylor gap (0.035 against 0.027) and Taylor fraction (0.83 against 0.77). These are the tail and nonlinearity facts on which the modulation acts.
- **Worse than the earlier run:** the Zumbach asymmetry (0.056 against 0.094, targets 0.103–0.129). Leverage is slightly too strong (−0.043, targets −0.031 to −0.040).

Per asset (seed 11): GPC 0.96, QCOM 1.28, INTC 1.18, LUV 1.32, NI 0.77. The high-volatility names (QCOM, LUV) are the hardest, because the grid-averaged mean volatility sits at the five-asset compromise.

7 Ablation

At $\theta^\star$ and its parameter grid, each quantity is set to a structural null (or the parameter is fixed at the mid-point of its two levels instead of gridded), and S_{avg} is recomputed on seed 11 (baseline 1.103). The 5-seed noise floor is $2\sigma = 0.156$.

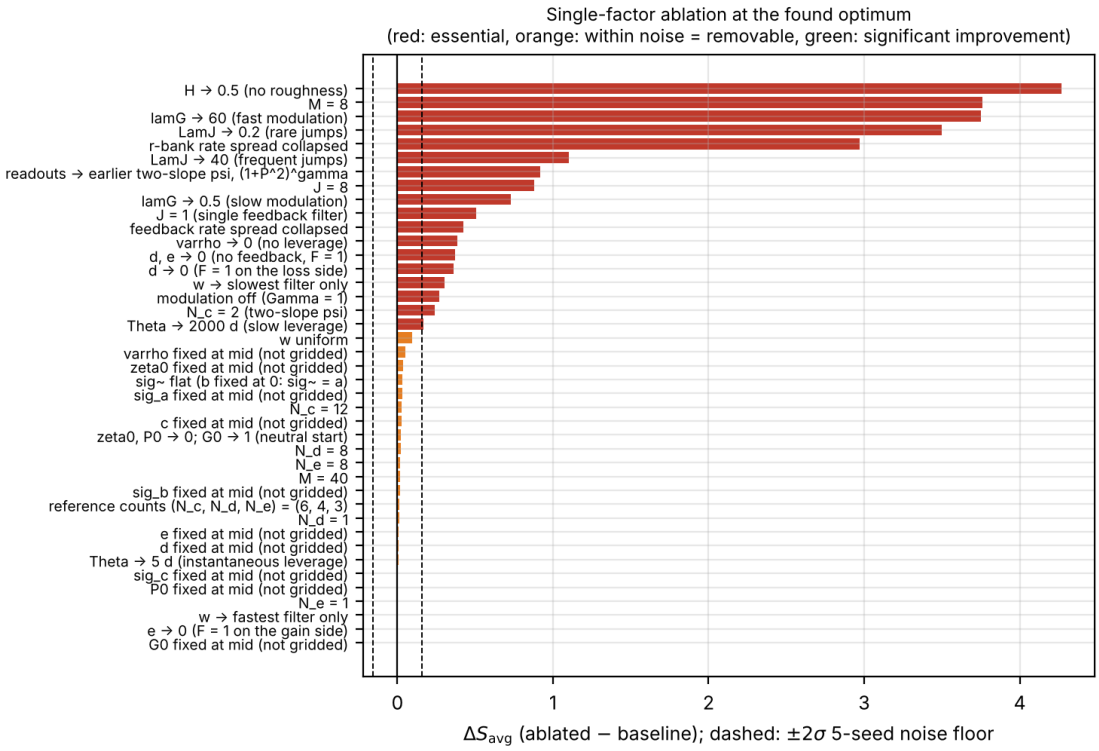


Figure 8: Single-factor ablation (41 ablations).

Table 5: Ablation, sorted by ΔS_{avg}. "$N_e = 1$" is $\theta^\star$ itself (identity check).

ablation	S_{avg}	Δ	verdict
$H \to 0.5$ (no roughness)	5.371	+4.268	essential
$M = 8$	4.862	+3.759	essential
$\lambda_\Gamma \to 60$ (fast modulation)	4.855	+3.752	essential
$\Lambda \to 0.2$ (rare jumps)	4.601	+3.499	essential
r-bank rate spread collapsed	4.074	+2.972	essential
$\Lambda \to 40$ (frequent jumps)	2.204	+1.101	essential
readouts $\to$ earlier two-slope ψ, $(1 + P^2)^\gamma$	2.020	+0.917	essential
$J = 8$	1.982	+0.879	essential
$\lambda_\Gamma \to 0.5$ (slow modulation)	1.834	+0.731	essential
$J = 1$ (single feedback filter)	1.611	+0.508	essential
feedback rate spread collapsed	1.528	+0.425	essential
$\varrho \to 0$ (no leverage)	1.489	+0.386	essential
$d, e \to 0$ (no feedback, $F \equiv 1$)	1.472	+0.370	essential
$d \to 0$ ($F \equiv 1$ on the loss side)	1.466	+0.363	essential
$w \to$ slowest filter only	1.406	+0.304	essential
modulation off ($\Gamma \equiv 1$)	1.373	+0.270	essential
$N_c = 2$ (two-slope ψ)	1.345	+0.242	essential
$\Theta \to 2000$ d (slow leverage)	1.270	+0.167	essential
w uniform	1.200	+0.097	within noise
ϱ fixed at mid (not gridded)	1.152	+0.049	within noise
ζ_0 fixed at mid (not gridded)	1.142	+0.039	within noise
$\tilde{\sigma}$ flat ($b \to 0$: $\tilde{\sigma} = a$)	1.136	+0.033	within noise
a fixed at mid (not gridded)	1.135	+0.032	within noise
$N_c = 12$	1.130	+0.028	within noise
c_m fixed at mid (not gridded)	1.129	+0.027	within noise
ζ_0, $P_0 \to 0$; $\Gamma_0 \to 1$ (neutral start)	1.125	+0.022	within noise
$N_d = 8$	1.124	+0.021	within noise
$N_e = 8$	1.119	+0.016	within noise
$M = 40$	1.119	+0.016	within noise
b fixed at mid (not gridded)	1.118	+0.015	within noise
reference counts (N_c, N_d, N_e) = (6, 4, 3)	1.117	+0.014	within noise
$N_d = 1$	1.117	+0.014	within noise
e_j fixed at mid (not gridded)	1.112	+0.009	within noise
d_j fixed at mid (not gridded)	1.112	+0.009	within noise
$\Theta \to 5$ d (instantaneous leverage)	1.109	+0.007	within noise
c (level) fixed at mid (not gridded)	1.107	+0.005	within noise
P_0 fixed at mid (not gridded)	1.103	+0.000	within noise
$N_e = 1$	1.103	+0.000	within noise
$w \to$ fastest filter only	1.103	−0.000	within noise
$e \to 0$ ($F \equiv 1$ on the gain side)	1.101	−0.002	within noise
Γ_0 fixed at mid (not gridded)	1.096	−0.007	within noise

1. **Rough bank.** Roughness (+4.27), the rate spread (+2.97) and $M \geq 16$ ($M = 8$: +3.76) dominate, as in the earlier run.

2. **Modulation (new layer).**

   - Switching it off ($\Gamma \equiv 1$) costs +0.27, so it is essential.
   - Its rates are sharply identified: fast modulation ($\lambda_\Gamma = 60$) costs +3.75, slow modulation (0.5) +0.73, rare jumps ($\Lambda = 0.2$) +3.50 and frequent jumps ($\Lambda = 40$) +1.10.
   - The time series select regimes of about a quarter, a few per year.

3. **Readouts (new shapes).**

   - Returning to the earlier two-slope ψ and $(1 + P^2)^\gamma$ costs +0.92, and a two-slope ψ alone ($N_c = 2$) costs +0.24: the piecewise-linear bounded readouts are a genuine improvement.

- The *numbers* of pieces beyond that are irrelevant. $N_c = 12$, $N_d \in \{1, 8\}$, $N_e = 8$ and the reference counts $(6, 4, 3)$ are all within 0.03.
- The loss side of F is essential ($d \to 0$: $+0.36$). The gain side is not ($e \to 0$: -0.002).

4. **Feedback bank.** $J = 1$ costs $+0.51$, $J = 8$ $+0.88$, a collapsed rate spread $+0.43$, and the slow filter alone $+0.30$. The fast filter alone is equivalent to $\theta^\star$.

5. **Leverage.** $\varrho \to 0$ costs $+0.39$, and a 2000-day window $+0.17$. A 5-day window is within noise.

6. **Parameters as grid dimensions, the initial state and the term structure.**

   - Fixing any parameter group at its mid-point instead of gridding it changes S_{avg} by at most 0.05.
   - A neutral start ($\zeta_0 = P_0 = 0$, $\Gamma_0 = 1$) costs $+0.02$, and a flat $\tilde{\sigma}$ costs $+0.03$.
   - Over 1342-day windows, the initial state and the term structure are invisible in the time-series facts.

8 Assessment

1. **Result.** With all 22 architecture hyperparameters free, the protocol reaches $S_{\text{avg}} = 1.103$ on the search seed and 1.215 ± 0.078 over five seeds. The earlier run of the same protocol, with the simpler model class, gave 0.991 and 1.066 ± 0.051. The difference, 0.15, is about 1.6 combined standard deviations. On the time-series facts, the richer class is therefore comparable to, not better than, the earlier class. It trades a better kurtosis and Taylor effect for a weaker Zumbach asymmetry.

2. **The new layers are needed within the class.** The modulation ($+0.27$) and the piecewise-linear readout shapes ($+0.92$ against the earlier readouts in this class) are essential. The counts N_c, N_d, N_e beyond a minimal resolution, the gain side of F, the term structure $\tilde{\sigma}(t)$ and the initial state are not.

3. **Architecture.** The rough bank found by the earlier run is confirmed ($H \approx 0.06$, $\lambda_{hi} \approx 235$), now with a long-memory lower rate. The feedback bank shrinks to one fast filter (about 3 days). The leverage window lengthens to about a month.

4. **The outer search.** For this model class, the protocol's bounded LM (stages A and C) stalls far from the optimum. The objective is piecewise constant in five integer hyperparameters and in Λ, and LM steps collapse on the step tolerance. A derivative-free coordinate sweep (stage D) is required. It lowered S_{avg} from 2.12 to 1.10 in about 200 evaluations.

5. **Limitations.**

   - One global design and one search seed.
   - Two hyperparameters end at a bound (λ_{lo}, κ_{lo}), which suggests widening the box.
   - The variogram fit is loose at $\theta^\star$.
   - The Zumbach asymmetry is under-reproduced.
   - The parameter levels, in particular for d and e, are protocol choices.

6. **Next steps.**

   - Wider λ_{lo} and κ_{lo} bounds.
   - A Zumbach-oriented driver of the feedback bank (for instance squared returns), since the feedback filter is the natural carrier of the Zumbach effect.

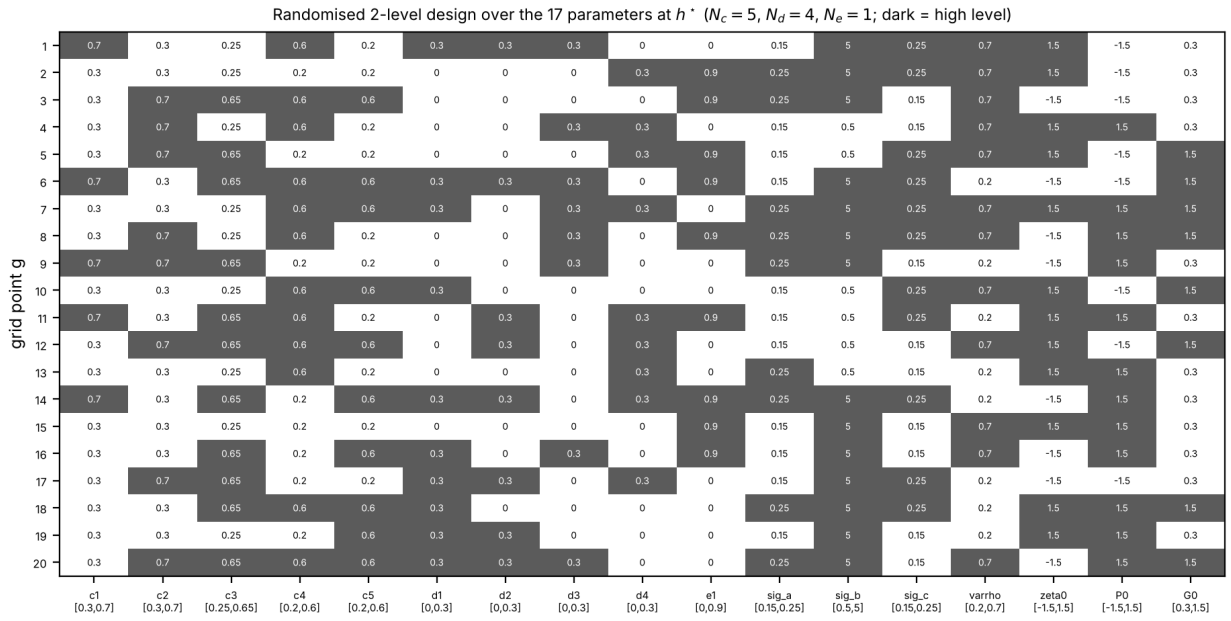


Figure 9: The randomised 2-level design of the 17 parameters at $\theta^\star$ ($N_c = 5$, $N_d = 4$, $N_e = 1$).

A Code reference

All files live in `code/` and carry the banner docstring. The only input is the data file, in the folder given by the environment variable `DATA_DIR`.

A.1 `gs_core.py`: model, simulation, statistics, score

Table 6: Constants of `gs_core.py`.

constant	meaning
`DT, ANN`	$\delta = 1/252$; $\sqrt{252}$
`T_CAL, N_WIN`	1342 days per window; 6 windows
`N_CAL, N_GRID, M_SUB`	8 paths per grid point; 20 grid points; 13 sub-steps per day
`ASSETS, STAT_NAMES, WEIGHTS, FLOORS`	the protocol's assets, statistics, score weights and tolerance floors
`NC_MAX, ND_MAX, NE_MAX`	12, 8, 8: numbers of slots for c, d, e (upper bounds of N_c, N_d, N_e)
`SCALARS, SCALAR_BOUNDS`	$(a, b, c, \varrho, \zeta_0, P_0, \Gamma_0)$ and their two levels
`C_NEG, C_POS, D_TOT, E_TOT`	slope levels left and right of 0; total amplitudes defining the d and e levels
`N_SLOTS, GRID_SEED`	35 slots; design seed 2026
`HYPS, HYP_BOUNDS, LOG_ENC, INT_HYPS`	the 22 hyperparameters, their box, the log-encoded subset, the integer subset

Table 7: Functions of `gs_core.py`.

function	role
`psi_knots(N_c)`	equispaced knots $-3 + 6k/N_c$
`F_knot_positions(N_d, N_e)`	knots $p_{-N_d}, \dots, p_{N_e}$ (Definition 2.1)
`interp_rows(x, knots, vals)`	row-wise linear interpolation (shared knots, per-path values), constant beyond the ends
`psi_values(c, N_c)`	knot values of ψ per path: cumulated $-c_m \Delta z$, shifted so that $\psi(0) = 0$
`F_values(d, e)`	knot values of F per path: 1+ reverse cumulative sums of d, and exp(−cumulative sums of e)
`sig_tilde(t, a, b, c)`	$\tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt})$
`load_assets()`	last 6×1342 days of returns and GK proxies per asset; GK/return-variance ratio
`gk_mean_factor, overnight_share`	$\mathbb{E}$ of the GK proxy of a unit Brownian session; per-asset ϕ and their mean
`stats_rows(x, v)`	the 11 statistics, row-wise (one row = one path or one window)
`score_refs(data)`	centre and tolerance per asset (protocol `make_score_ref`)
`encode, decode, enc_bounds`	search coordinates $\leftrightarrow$ hyperparameter dict (integers rounded); box in encoded coordinates
`build_bank(h)`	λ, q, γ_0, $u(\Theta)$, κ, w, $(\lambda_\Gamma, \Lambda, a_J)$, readout knots, readout switch
`param_bounds(N_c, N_d, N_e)`	the two levels of every active parameter (Table 1)
`slot_of(name)`	slot of a parameter in the 35-column bit matrix
`param_grid(N_c, N_d, N_e, ..., fixed)`	the slot design (Section 4.3); `fixed` pins a parameter, or a whole group `c`/`d`/`e`, to a value or to `"mid"` (ablation)
`grid_arrays(G, ...)`	stacks the c, d, e columns
`simulate(bank, G, ...)`	vectorised exact scheme of Section 5, from the initial state; returns x, v (and σ, Γ on request)
`averaged_stats`	statistics averaged over the 8 paths, then over the grid points
`score_from_stats`	55-entry residual, S_{avg}, per-asset scores
`evaluate(h, refs, phi, ...)`	Algorithm 1

`simulate`, step by step.

1. Expand the design: each of the 20 rows is repeated for its 8 paths (160 rows). Build the per-path knot values of ψ and F.
2. Precompute the sub-step constants: $a_i = e^{-\lambda_i h}$, the exact OU/Brownian correlation k_c, $e^{-\kappa_j h}$, $e^{-\lambda_\Gamma h}$, $1 - e^{-\Lambda h}$.
3. Set $\Gamma = \Gamma_0$, r_0 by (2.6) (Lemma 5.1) and $s_0 = P_0 w$.
4. For each of the 1342 days and each of its 13 sub-steps, apply items 1–6 of Section 5. Open, high and low are recorded from the first session sub-step.
5. Return the daily returns and GK proxies, with the proxy floored at 10^{-3} times the path median.

Note. `h["modul"]=False` switches the modulation off ($\Gamma \equiv 1$). `h["readout"]="linear"` replaces $e^{\psi} F$ by the earlier readouts at the mid-points of their earlier boxes: $\psi(z) = -0.5z$ for $z < 0$, $-0.4z$ for $z > 0$, and $(1 + P^2)^{0.075}$. Both switches are used only by the ablation; the search always runs the full model.

A.2 Scripts

script	role
`gs_search.py`	stage A (cold start) and stages C (from the best logged point). Functions: `cold_start`; `resid`, which writes one trace row per call; `make_parallel_jac`, forward differences with 22 columns on `NWORK` processes. Usage: `[NWORK=k] python gs_search.py 15`
`gs_polish.py`	stage D, the greedy coordinate sweep. Usage: `NWORK=2 python gs_polish.py 2 7` (passes, values per coordinate)
`gs_global.py`	stage B, the Latin hypercube. Usage: `python gs_global.py 48 7`; `TRACE` selects the trace file
`gs_post.py`	`repeat` (5 seeds and per-grid-point statistics), `heatmap` (4×4 in (M, J)), `ablation` (41 ablations)
`export_json.py`	targets, measurement constants, design, labelled trace
`gs_figs.py`	Figures 1–9
`reproduce.sh`	the full pipeline

Note. Reproducibility. The trace is the single source of truth. Every evaluation of every stage appends one JSON row: encoded θ, decoded h, Q, S_{avg}, per-asset scores, the 11 averaged statistics, NNLS fit error and elapsed time. Stage C restarts from the row with the smallest Q, so an interrupted run resumes. With `NWORK`> 1 the Jacobian rows are appended in completion order; the stage boundaries come from the leg counts in `legs.json`.

B Results files

- `targets.json`: centres, tolerances and window values per asset; statistic names, weights, floors.
- `measurement.json`: ϕ per asset and the value used.
- `design.json`: slot layout and levels, the parameter grid at the optimum's readout counts, hyperparameter box and encoding, numerical constants.
- `trace.json`: all evaluations, each with its stage label.
- `legs.json`, `search_result.json`: leg sizes; result of the last TRF leg.
- `post_repeat.json`, `post_heatmap.json`, `post_ablation.json`: post-search analyses.